# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 8 — SCIENCE CHAPTER TEST SET B

Chapter 9: **The Amazing World of Solutes, Solvents and Solutions** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** A substance has mass 270 g and volume 100 cm³. Its density is: 1

**(a)** 2.7 g/cm³ **(b)** 5.4 g/cm³ **(c)** 0.37 g/cm³ **(d)** 1.35 g/cm³

**2.** In a salt–water solution, the solute is: 1

**(a)** air **(b)** salt **(c)** container **(d)** water

**3.** A solution in which no more solute dissolves at that temperature is: 1

**(a)** dilute only **(b)** unsaturated **(c)** pure water **(d)** saturated

**4.** A cold soft drink stays fizzy longer than a warm one because: 1

**(a)** gas dissolves better in cold liquid **(b)** gas dissolves better in hot liquid

**(c)** sugar is absent **(d)** cold water has no gas

**5.** Mass is measured with a: 1

**(a)** ruler **(b)** weighing balance

**(c)** measuring cylinder **(d)** thermometer

**6.** The SI unit of density is: 1

**(a)** kg/m³ **(b)** kg **(c)** m³ **(d)** N/m²

**7.** Ice floats on water because ice has: 1

**(a)** lower density than water **(b)** more mass

**(c)** no particles **(d)** higher density than water

**8.** Pressure has the strongest effect on the density of: 1

**(a)** solids **(b)** metals **(c)** gases **(d)** liquids

**9.** Which of the following is a solid–liquid solution? 1

**(a)** air **(b)** salt in water **(c)** soda water **(d)** brass

**10.** An unsaturated solution becomes saturated when: 1

**(a)** it is stirred

**(b)** water is added

**(c)** more solute is added until it stops dissolving

**(d)** it is poured

## SECTION — B (Very Short Answer Type) 4 × 2 = 8

*Answer in about 10–15 words each.*

**11.** How does temperature affect the solubility of solids and gases in liquids? 2

**12.** What is a meniscus? How is the reading taken? 2

**13.** Define density and write its formula. 2

**14.** Write two units of density. 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 30–40 words each.*

**15.** Wood of density 0.6 g/cm³ and iron of density 7.8 g/cm³ are placed in water. Predict and explain. 3

**16.** Explain how pressure affects density of gases and liquids. 3

**17.** Describe how to measure the volume of an irregular stone using a measuring cylinder. 3

OR

Explain why the egg floats in salt solution but sinks in plain water. 3

**18.** Explain the term expansion in the context of water freezing. 3

OR

List three factors that affect how quickly a solute dissolves. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 80–100 words each.*

**19.** Explain why ice floats and why this is vital for life in lakes during winter. 5

OR

Describe how density depends on temperature and pressure with examples. 5

**20.** A solid block has mass 480 g and volume 160 cm³. 5

(i) Find density.

(ii) Will it float in water?

(iii) Find mass of 50 cm³ of it.

OR

Explain with an activity how a raw egg can be made to float in water. 5

## END OF QUESTION PAPER — Total: 40 Marks

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